

Madhan Karthikeyan

+91 93619 51624 | madhankrthik@gmail.com | madhan-karthikeyan.vercel.app | linkedin.com/in/madhan-karthikeyan | github.com/madhan-karthikeyan

SUMMARY

Computer Science undergraduate (VIT + IIT Madras) with hands-on internship and project experience across ML systems and full-stack backend engineering — computer vision, embedding-based retrieval, time-series forecasting, REST API design, CLI tooling, and async task pipelines; skilled at building end-to-end systems from data curation through optimized production deployment.

EDUCATION

Vellore Institute of Technology (VIT) <i>B.Tech in Computer Science & Engineering — CGPA: 8.80/10</i>	Vellore, Tamil Nadu Aug 2023 – Present
Indian Institute of Technology, Madras <i>B.S. in Data Science and Applications</i>	Online Sep 2023 – Present

WORK EXPERIENCE

AI Engineer Intern <i>National Informatics Centre Govt. of India</i>	May 2025 – Jul 2025 Chennai, Tamil Nadu
- Architected a triage-to-resolution pipeline integrating a vision-language model for structured issue extraction with a vector similarity index over 150+ guideline documents (4 GB), cutting resolution time from >24h to under 1 minute. - Engineered the document ingestion service with chunking, embedding generation, and metadata-filtered cosine retrieval, enabling sub-second semantic search over 4 GB without dedicated search infrastructure.	
Computer Vision Intern <i>GloballyGI</i>	Jan 2026 – Jun 2026 Vellore, Tamil Nadu
- Engineered a multi-object food detection pipeline (YOLOv8n + D-FINE), achieving 80–85% mAP across 50 food classes on a custom-curated 5,000-image annotated dataset. - Designed class-balancing and targeted augmentation pipelines for 50 categories, reducing per-class mAP variance by ~35% and improving generalization on occluded, multi-object scenes. - Reduced inference latency by ~25% through lightweight architecture selection and post-training optimization, enabling real-time detection suitable for production deployment.	

PROJECTS

Apex Sprint Planner <i>Explainable Sprint Planning Platform</i>	FastAPI, React, TypeScript, SQLAlchemy, Celery, Redis, MinIO, Docker, GitHub Actions GitHub
- Built a full-stack sprint planning platform with a FastAPI backend, async SQLAlchemy + Alembic data layer, and a React/TypeScript/Vite frontend spanning dashboard, dataset upload, plan generation, explainability, and kanban approval views. - Designed an async planning pipeline (Celery + Redis) executing five stages — context extraction, weight learning, ILP-based constrained optimization (PuLP/CBC), and per-story explainability generation — with live job-status polling via REST. - Containerized the full stack with Docker Compose (backend, frontend, Postgres, Redis, MinIO, Celery worker) and shipped a GitHub Actions CI workflow with automated pytest suites across root and backend test directories.	
DecisionDrift — ADR Governance CLI <i>Deterministic Decision Governance System</i>	Python, Click, AST/Tree-sitter, GitHub Actions, PyPI GitHub
- Architected a deterministic rule engine compiling ADR prohibitions into 5 rule classes (dependency, import, path, API, config), enforced via AST-based static analysis on diffs and full-repo scans; validated against a 21-patch labeled benchmark achieving 95.2% Recall@5 and 85.7% Recall@1. - Published a PyPI package and a GitHub Action (live on GitHub Marketplace) with an ADR bootstrap engine, pre-commit guard hook, and optional LLM semantic reviewer; engineered Tree-sitter impact-analysis across Python, JS, TS, Go, Java, and Rust.	
JobLens — Personal Placement Operating System <i>Full-Stack Job Search Automation Platform</i>	FastAPI, PostgreSQL, SQLAlchemy (async), React, TypeScript, Docker GitHub
- Built a multi-source aggregation pipeline ingesting listings from Adzuna, JSearch, YC, and 4 ATS platforms (Greenhouse, Lever, Ashby, Workday) into an async PostgreSQL store (SQLAlchemy + Alembic), orchestrated via daily cron jobs. - Designed a deterministic ranking engine scoring opportunities on role alignment (35%), skill match (30%, 90+ skills / 46 aliases), behavior affinity (20%, feedback-learned), and freshness decay (10%), returning a score breakdown and match reasons per listing. - Delivered a FastAPI backend and React + Vite + TypeScript frontend (job board, kanban, skill-gap analysis) with an automated Discord digest webhook, fully containerized via Docker.	
MediCore — Hospital Management Platform <i>Full-Stack Distributed System</i>	Flask, Vue.js, Celery, Redis, PostgreSQL, Razorpay, JWT
- Architected a multi-role hospital platform on Flask + Vue.js with JWT-secured RBAC across ~40 REST APIs, supporting independent admin, doctor, and patient workflows across a 3-tier user hierarchy. - Designed async task pipelines using Celery + Redis for reminder dispatch, report generation, and bulk data export, decoupling long-running jobs from the HTTP request cycle and eliminating request timeout failures.	

- Implemented event-driven Redis cache invalidation maintaining **<150ms p95 API latency** under concurrent multi-role load; integrated Razorpay with idempotent transaction handling to eliminate duplicate payment risk.

Satellite Image Search System

RemoteCLIP, ChromaDB, DBSCAN, FastAPI, React, Cesium

Embedding-Based Geospatial Retrieval

GitHub

- Built a semantic retrieval system over 10,000+ geo-tagged satellite images using RemoteCLIP embeddings and ChromaDB vector search, achieving **<500ms end-to-end query latency** across multispectral datasets.
- Selected RemoteCLIP over vanilla CLIP for domain-specific pretraining on 4.7M+ remote sensing image-text pairs, yielding higher retrieval precision on multispectral and SAR queries.
- Applied DBSCAN with Haversine metric to cluster results into geographic regions; exposed retrieval via FastAPI with bounding-box queries and 3D visualization using Cesium ion.

Cinema Audience Forecasting System

LightGBM, XGBoost, CatBoost, Tweedie Regression, FastAPI, Python

Time-Series Ensemble Forecasting

GitHub

- Built a 3-model gradient boosting ensemble (LightGBM + XGBoost + CatBoost) with Tweedie regression objective to model zero-inflated audience counts across 112 theaters, achieving **RMSE 21.6 / R² 0.54** on a held-out validation window.
- Engineered lag (7/14/21/28-day), rolling-window, and cyclic calendar features with TimeSeriesSplit cross-validation to prevent leakage; deployed a lag-blend inference pipeline ($\alpha = 0.2$) with graceful degradation (3-model → 2-model → lag fallback) via FastAPI.

Cluster-Aware Semantic Retrieval & Caching System

GMM, FAISS, FastAPI, Python

ML Retrieval System

GitHub

- Partitioned the embedding space using Gaussian Mixture Models to scope FAISS search within cluster regions, reducing average query latency by **~30%** over naïve global search.
- Built a cluster-indexed semantic cache for result reuse on similar queries; the full pipeline (embed → GMM route → cache → FAISS fallback) sustained a **~35% cache hit rate** under repeated-query workloads, served via FastAPI.

Responsible Financial AI — RAG System

LlamaIndex, ChromaDB, FastAPI, ReactJS

Full-Stack Retrieval-Augmented Generation

- Built a production-grade RAG pipeline using LlamaIndex and ChromaDB for context-grounded financial Q&A, with structured document ingestion, embedding indexing, and cosine similarity retrieval to reduce hallucinations.
- Delivered a FastAPI backend and ReactJS frontend serving explainable, source-cited responses with **<2s end-to-end latency** across a 1,000+ document knowledge base.

DriftBench-TS — Time-Series Benchmarking Framework

Python, Flask, React, LightGBM, Configuration-Driven Pipelines

ML Systems + Software Engineering

- Built a configuration-driven benchmarking framework supporting 5+ forecasting algorithms and 8+ concept drift detectors with plug-and-play integration, rolling-window simulation, and automated retraining strategies.
- Exposed experiment orchestration via a Flask REST API and React dashboard providing real-time metric visualization across 6 benchmark datasets, reducing per-experiment setup time from hours to minutes.

SKILLS

ML & Modeling: PyTorch, Scikit-learn, LightGBM, XGBoost, CatBoost, YOLOv8, Time-Series Forecasting, Feature Engineering, Tweedie Regression

CV & Retrieval: CLIP / RemoteCLIP, D-FINE, FAISS, ChromaDB, Vector Search, Embedding Pipelines, Object Detection

ML Systems: RAG Pipelines, GMM Clustering, DBSCAN, Semantic Caching, LlamaIndex

Backend: Python, FastAPI, Flask, REST API Design, JWT Auth, Async I/O (SQLAlchemy, asyncpg), Click (CLI), Celery, Redis

Frontend: React, Vite, TypeScript, Vue.js, Tailwind CSS

Databases: PostgreSQL, SQLite, MongoDB, ChromaDB, Alembic Migrations

DevOps & Systems: Docker, Docker Compose, GitHub Actions (CI/CD), AWS (EC2, S3), MinIO, AST & Static Analysis, PyPI Packaging

Programming & Data: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, JavaScript / TypeScript

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate | IIT Madras: Advanced Certificate in Machine Learning & Data Science